TECHNICAL DILIGENCE REVIEW - RECORD 018

Helioscope Robotics, Inc. - diligence file reference TECHNICAL_DILIGENCE-018

1. Terms of reference, method and interview programme

Report TDR-2026-16 was prepared by Pinehurst Systems Assurance for Ardent Peak and covers the reliability record, incident history and post-incident follow-through of the fleet platform at Helioscope Robotics, Inc. as at 2026-04-27. As instructed, Pinehurst Systems Assurance confined report TDR-2026-16 to the reliability record, incident history and post-incident follow-through of the fleet platform; three adjacent questions raised during the work are listed as follow-ups rather than answered here. For the transaction, the present report identifies 3 matters the reviewer would raise before closing and 8 for the post-close engineering plan. The costed remediation implied by report TDR-2026-16 is $1,355,000 over the coming four quarters, which is within the ordinary engineering budget. The company reviewed a draft of this report for factual accuracy and its corrections are reflected without altering any conclusion.

R. Ostrowski was granted read access to the code, the pipeline and the incident tracker for the purposes of this report; production access was not requested. R. Ostrowski led the work behind TDR-2026-16, combining document review, 9 interviews and hands-on inspection of running systems. R. Ostrowski chose the sample for report TDR-2026-16 after reading the dependency map, so the selection is not proportionate to headcount or to code volume. 7 sites contributed evidence to this report, and one further site was in acceptance testing and was excluded. R. Ostrowski has stated the limitations at the point they bite rather than collecting them into a disclaimer.

Interview evidence in TDR-2026-16 came from 9 sessions across 6 teams, each minuted and circulated for correction. Pinehurst Systems Assurance asked the same four questions of every team interviewed for the report at TDR-2026-16, which is how the inconsistencies below were found. R. Ostrowski tested a sample rather than the population in TDR-2026-16, and the sampling basis is stated in each section. The report at TDR-2026-16 did not rely on management representation alone; every material claim was traced to a document, a measurement or an observation. R. Ostrowski has not smoothed the sequence of this report; it follows the order in which the evidence arrived.

The convention throughout the report at TDR-2026-16 is that an absence of evidence is reported as such rather than as evidence of absence. R. Ostrowski treats a backlog ticket as evidence of awareness and a merged change with a test as evidence of remediation, and this report keeps the two apart. TDR-2026-16 does not include a code audit; the reviewer read code where a specific claim required it and no further. R. Ostrowski did not visit the two most recently commissioned sites, so the commissioning quality there is taken from the company's records alone. Pinehurst Systems Assurance has reported the position as at 2026-04-27 and has not attempted to predict where it will be at completion.

Two document requests remained partly unsatisfied when the present report was delivered, and the affected conclusions are marked. The reviewer notes that the present report was prepared while three remediation programmes were in flight, so several positions will have moved by completion. The report at TDR-2026-16 recommends that 3 items be reflected in the disclosure schedules and that the remainder be tracked in the post-close plan. None of the findings in report TDR-2026-16 would in the reviewer's view justify withholding completion, and two would justify a specific undertaking. The numbering in the report at TDR-2026-16 follows the order of the work rather than the order of importance.

The sample behind this report is 14 services and 7 sites; the selection was the reviewer's, not the company's. Coverage of the report at TDR-2026-16 is deliberately uneven: the safety and control path was examined closely and the reporting path only at a high level. Grading in this report is deliberately coarse, because a finer scale would imply a precision the underlying evidence does not carry. The present report distinguishes three positions: verified defect, credible concern, and matter requiring further work. The third category is not empty. Report TDR-2026-16 does not restate the company's own reporting where the reviewer was able to verify the underlying data instead.

Schedule I - Interview programme and follow-up requests arising (TDR-2026-16)

Date	Participant	Role	Principal topic	Follow-ups raised	Follow-ups open
2026-10-27	Erik Palmgren	principal engineer	Robot Shadow Store	3	0
2025-12-03	Erik Palmgren	engineering owner	Video Snippet Service	1	0
2026-03-08	Yuki Harada	site reliability lead	Billing Usage Meter	5	0

Date	Participant	Role	Principal topic	Follow-ups raised	Follow-ups open
2025-12-03	Priya Raghunathan	engineering owner	Map Authoring Studio	1	1
2026-06-02	Wanda Njoku	security engineer	Billing Usage Meter	2	1
2026-07-18	Sam Oyelaran	head of engineering	Video Snippet Service	4	4
2025-12-03	Nadia Semenova	team lead	Traffic Manager	1	1
2025-08-27	Colin Achebe	engineering owner	Robot Shadow Store	2	2
2026-07-24	Marta Kellerman	quality lead	Gripper Controller	4	2
2025-08-29	Tomas Villareal	security engineer	Identity and Authorization	2	1
2026-04-23	Yuki Harada	release manager	Lidar Driver	5	3
2026-02-05	Colin Achebe	engineering owner	Charge Scheduler	5	2
2026-06-09	Yuki Harada	field reliability lead	Fleet Health Monitor	5	3
2026-02-15	Marta Kellerman	engineering owner	OTA Updater	3	1
2026-05-30	Sam Oyelaran	principal engineer	Event Bus	2	0
2026-08-23	Farida Cisse	engineering owner	Payload Scale Interface	4	1
2026-04-27	Rahul Menon	engineering owner	Simulation Harness	4	4
2026-02-12	Marcus Duplessis	release manager	Billing Usage Meter	1	1
2026-01-27	Priya Raghunathan	team lead	Hardware-in-the-Loop Rig	2	1
2026-05-20	Marta Kellerman	head of engineering	Battery Management Service	5	1
2026-01-28	Lucia Ferrante	field reliability lead	OTA Updater	3	1
2026-04-07	Sam Oyelaran	team lead	Charge Scheduler	1	0
2026-01-08	Marta Kellerman	engineering owner	Hardware-in-the-Loop Rig	1	1
2026-06-07	Rahul Menon	team lead	Fleet Allocator	2	2
2025-08-28	Colin Achebe	field reliability lead	Config Distribution	3	2
2026-09-21	Nadia Semenova	head of engineering	Motion Planner	1	1
2026-01-08	Marcus Duplessis	head of engineering	Traffic Manager	5	3
2025-12-06	Erik Palmgren	head of engineering	Motion Planner	5	5
2025-08-24	Nadia Semenova	security engineer	WMS Connector	1	1
2025-11-05	Bettina Vogel	release manager	Site Provisioning Service	4	0

2. INC-2026-062 and the incident population examined

On 2026-06-02 the record for incident INC-2026-062 opens: an emergency release path with a single approver was used for a change that was not an emergency. As to whether the INC-2026-062 remediation actually landed: not landed; the change is scoped but not scheduled. The blast radius of INC-2026-062 was no production site and no robot directly, the exposure being to internal systems. Colin Achebe committed after INC-2026-062 to require a justification and a second approver within one working day for every emergency release, due 2026-08-31. The classification applied to process control event INC-2026-062 is process control, and the affected component is Release Orchestrator. process control incident INC-2026-062 was verified to the extent the available evidence permits, and the residual uncertainty is stated rather than absorbed.

Assessed on its own record, process control event INC-2026-062 shows that the blast radius was set by a deployment topology decision made years earlier. Beyond the direct cause, incident INC-2026-062 shows that the policy defines an emergency but the tool does not ask which condition applies. Not applicable; process only elapsed before process control incident INC-2026-062 was mitigated, with resolution at open at the review date. Release Orchestrator incident INC-2026-062 reached the engineering team when the release audit found it during the quarterly review. Incident INC-2026-062 traces back to a single fact: the emergency path is available to any release engineer without a justification field. process control incident INC-2026-062 is reported at the evidence strength available and has not been promoted to fit the narrative.

Strong on the cause and limited on the contributing factor is the evidence grade for Motion Planner incident INC-2025-103, and the reader should not read severity into it. The tie-break landed on 2025-10-19; the scenario was reinstated on 2025-09-02 is the position on the algorithm defect incident INC-2025-103 commitment as at the review date. The affected population in Motion Planner incident INC-2025-103 was one site and 2 robots. Drawing the evidence in Ironwood Grocery Group incident INC-2025-103 together, the elapsed resolution time reflects a scheduling decision rather than an engineering difficulty. Incident INC-2025-103 was caused by a design gap: the yield heuristic had no tie-

break and both robots computed the same deference score. INC-2025-103 appears twice in the working papers for TDR-2026-16 because the first answer was corrected after a follow-up request.

Incident INC-2025-103 was detected because the queue depth alert introduced after the March incident fired correctly. Remediation for Motion Planner incident INC-2025-103 is to add a deterministic tie-break and return the pinch-point scenario to the nightly suite; Priya Raghunathan owns it and the committed date is 2025-10-31. Algorithm defect incident INC-2025-103 classifies as a algorithm defect event against Motion Planner. 2025-08-22 is the date of Motion Planner incident INC-2025-103, on which two robots repeatedly yielded to each other in a narrow aisle and both stalled for nineteen minutes. INC-2025-103 took 19 m to stabilise and 58 days to close out. algorithm defect incident INC-2025-103 rests on a single source, and the reviewer has said so rather than implying corroboration.

A contributing factor in modelling incident INC-2026-057 was that the two affected sites moved to opportunity charging after the estimator had been validated. INC-2026-057 is recorded as follows on delivery: not landed; data collection is running and the imbalance alert is in review. So far as reach is concerned, INC-2026-057 involved 2 sites and 5 robots. Peter Halvorsen committed after INC-2026-057 to refit the estimator on opportunity-charge data and add an imbalance alert independent of state of health, due 2026-09-30. The two durations recorded for Dunbar Industrial incident INC-2026-057 are not a service outage; robots derated to mitigate and open at the review date to resolve. incident INC-2026-057 was compared with the equivalent position at two other components, and it is neither the best nor the worst.

Battery Management Service accounts for modelling incident INC-2026-057, a modelling event. In modelling incident INC-2026-057 the first signal was a robot derated mid-shift and the pack was pulled for analysis, and the reviewer treats that as reported detection. Modelling incident INC-2026-057 began on 2026-05-19 when state-of-health estimates on opportunity-charged packs drifted enough to hide a cell imbalance until derating. Root cause in modelling event INC-2026-057 was that the estimator was fitted on packs that follow a full charge cycle. Grading Battery Management Service incident INC-2026-057 on evidence rather than on consequence gives moderate, resting on the team's own post-mortem and one supporting log extract. modelling incident INC-2026-057 is included because a reader of TDR-2026-16 should be able to disagree with the inference without repeating the work.

Reported detection applies to process gap incident INC-2025-096, since discovered the next morning when the alarm was found unacknowledged. Incident INC-2025-096 records that on 2025-08-05, an overnight alarm routed to a shared mailbox because the on-call rotation had a two-hour gap. Internal incident INC-2025-096 touched no production site and no robot directly, the exposure being to internal systems, and internal was the affected account. Classifying INC-2025-096 as process gap matters because the Alarm Router backlog contains two further items of the same kind. Process gap event INC-2025-096 rests on limited; the log retention window had closed before the reviewer asked for the raw data evidence, which the reviewer states separately from the severity of the finding. process gap event INC-2025-096 is not material on its own and belongs in this report because of what it reveals about the surrounding process.

The reviewer treats one contributing factor in INC-2025-096 as the more important lesson: the rotation source of truth was a spreadsheet maintained outside the alarm router. The two durations recorded for incident INC-2025-096 are not applicable; no service impact to mitigate and 26 days to resolve. Root cause in process gap event INC-2025-096 was that the routing table fell back to a mailbox rather than to a secondary responder. Process gap event INC-2025-096 is recorded as follows on delivery: the fallback change landed on 2025-08-31; the rotation import was deferred to 2026. What incident INC-2025-096 tells the reviewer is that the customer-reported detection path is the finding, not the defect itself. Alarm Router incident INC-2025-096 was checked against the company's own register, which had not been updated since the change landed.

Incident INC-2026-004 records that on 2026-01-13, relocalisation after a lift-and-place exceeded the operator prompt timeout at a newly surveyed site. Performance event INC-2026-004 is recorded as follows on delivery: not landed; the matcher replacement is in design and the re-survey is scheduled. Incident INC-2026-004 carries limited, being a single account corroborated only by the ticket timeline evidence, and the reviewer declined to raise the finding beyond what that supports. Following INC-2026-004 the team undertook to replace the fallback matcher and re-survey the affected aisles, with Rahul Menon accountable by 2026-06-30. The two durations recorded for performance incident INC-2026-004 are not a service outage; 5 days to a workaround to mitigate and open at the review date to resolve.

performance incident INC-2026-004 was raised again with the owning engineer after the initial interview, and the position did not change.

The engineering team's root cause for incident INC-2026-004 is that the fallback scan matcher is slower than the primary and is selected whenever the map cache misses, and the reviewer found the analysis sound. Performance is the class of Kestrel Medical Supply incident INC-2026-004; Localization Engine is the component. INC-2026-004 is not read as an isolated failure: the elapsed resolution time reflects a scheduling decision rather than an engineering difficulty. Beyond the direct cause, performance incident INC-2026-004 shows that the site was surveyed with a sparse feature set that increases cache misses. Kestrel Medical Supply incident INC-2026-004 was detected because the site's commissioning engineer raised it during acceptance testing. performance incident INC-2026-004 would move from a concern to a verified defect if one further measurement were supplied.

Remediation for INC-2025-139 is to split control traffic onto a dedicated topic with its own quota and revisit the topic inventory quarterly; Erik Palmgren owns it and the committed date is 2026-01-31. Event Bus accounts for incident INC-2025-139, a capacity event. Assessed on its own record, incident INC-2025-139 shows that the failure was in the detection design rather than in the code that failed. In capacity event INC-2025-139, the topic inventory had recorded the sharing as acceptable in 2023 and it was never revisited, which is a process finding rather than a code finding. Capacity incident INC-2025-139 carries strong, since the reviewer reproduced the failure mode in a laboratory environment evidence, and the reviewer declined to raise the finding beyond what that supports. multiple sites incident INC-2025-139 has an owner who accepted the finding immediately, which is worth recording alongside the finding itself.

Capacity event INC-2025-139 was raised on 2025-11-02 after a telemetry backlog on a shared topic delayed control events by up to eleven seconds. On delivery, INC-2025-139 shows this: the split landed on 2026-01-04; the quarterly review has not yet run. Multiple sites felt incident INC-2025-139, which affected 7 sites and 246 robots. What went wrong in INC-2025-139 was that control and telemetry traffic shared one topic and one consumer group budget. Mitigation of capacity incident INC-2025-139 took 2 h 47 m and full resolution took 63 days. Event Bus incident INC-2025-139 is reported here in full because a summary of it in an earlier draft of TDR-2026-16 was misread as an assurance.

Resource leak incident INC-2025-041 closure status is landed on 2025-05-22, verified in the reservation replay suite. Underneath Crestline Fulfillment incident INC-2025-041 lies the design decision that the reservation lease had no expiry, so release depended on the departing robot. The contributing factor recorded against INC-2025-041 is that the alert rule for queue depth existed but was scoped to a metric the site did not emit. Incident INC-2025-041 was raised on 2025-03-14 after a robot lost its network link mid-traverse and its corridor reservation was never released. 1 h 52 m elapsed before Crestline Fulfillment incident INC-2025-041 was mitigated, with resolution at 9 days. Traffic Manager incident INC-2025-041 would move from a concern to a verified defect if one further measurement were supplied.

In taxonomy terms resource leak incident INC-2025-041 is a resource leak incident in Traffic Manager. The judgement recorded against Traffic Manager incident INC-2025-041 is that the remediation addresses the immediate cause and leaves the contributing factor in place. Traffic Manager incident INC-2025-041 reached the engineering team when a site supervisor telephoned the on-call engineer after robots queued at one intersection. The remediation committed after Crestline Fulfillment incident INC-2025-041 was to give every corridor reservation a lease with an expiry and add a site-level queue depth alert, owned by Owen Brantley with a due date of 2025-05-30. So far as reach is concerned, Crestline Fulfillment incident INC-2025-041 involved 2 sites and 61 robots. resource leak incident INC-2025-041 was cross-checked against the engineering record supplied for TDR-2026-16, and the two agree.

Reported detection applies to Remote Operations Console incident INC-2025-078, since an internal engineer noticed the discrepancy during a demonstration. INC-2025-078 affected 4 site or tenant with no robot taken out of service. 1 h 15 m elapsed before Harborline Systems incident INC-2025-078 was mitigated, with resolution at 34 days. Remote Operations Console owns INC-2025-078, which the reviewer classes as stale state. In Harborline Systems incident INC-2025-078, no test covered a roster change during an open operator session, which is a process finding rather than a code finding. INC-2025-078 was verified to the extent the available evidence permits, and the residual uncertainty is stated rather than absorbed.

The reviewer checked the stale state event INC-2025-078 commitment against the code and release records; landed on 2025-07-28; the regression test was still marked flaky at review. On 2025-06-24 the record for Remote Operations

Console incident INC-2025-078 opens: the console listed robots that had been removed from a site and an operator commanded a decommissioned unit. The post-mortem for stale state event INC-2025-078 identifies the cause as follows: the session list cached the site roster and only invalidated it on a page reload. The reviewer's reading of Harborline Systems incident INC-2025-078 is that the same class of defect can recur in two sibling components that were not changed. The commitment register shows Remote Operations Console incident INC-2025-078 against an undertaking to invalidate the roster cache on a fleet membership event and add a session-level regression test, with Anjali Sekhon named and 2025-08-31 committed. stale state event INC-2025-078 sits inside the sample rather than being an exception to it, which is the reason it is reported at length.

Schedule F - Incident population examined, with follow-through status (TDR-2026-16)

Incident	Date	Component	Sites affected	Robots affected	Time to mitigate	Time to resolve	Root cause class	Remediation landed
INC-2025-041	2025-03-14	Traffic Manager	2	61	1 h 52 m	9 days	resource leak	landed
INC-2025-047	2025-04-02	Lidar Driver	1	12	38 m	3 days	timing	partly landed
INC-2025-052	2025-04-21	Charge Scheduler	1	44	4 h 10 m	16 days	scheduling	landed
INC-2025-058	2025-05-09	WMS Connector	2	0	2 h 35 m	21 days	data integrity	landed
INC-2025-063	2025-05-27	Fleet Allocator	3	28	6 h 05 m	12 days	policy defect	landed
INC-2025-071	2025-06-11	Telemetry Ingest	6	214	52 m	52 m	availability	landed
INC-2025-078	2025-06-24	Remote Operations Console	4	0	1 h 15 m	34 days	stale state	landed
INC-2025-084	2025-07-08	OTA Updater	1	11	27 m	5 days	state persistence	landed
INC-2025-090	2025-07-19	Safety Controller	1	3	immediate stop, 22 m to return to service	41 days	hardware fault	partly landed
INC-2025-096	2025-08-05	Alarm Router	0	0	not applicable; no service impact	26 days	process gap	partly landed
INC-2025-103	2025-08-22	Motion Planner	1	2	19 m	58 days	algorithm defect	partly landed
INC-2025-110	2025-09-03	Analytics Warehouse Loader	1	0	not applicable; reporting only	18 days	data integrity	landed
INC-2025-118	2025-09-17	Identity and Authorizatio n	0	0	3 h 40 m to revoke and rotate	7 days	secret exposure	partly landed
INC-2025-124	2025-10-01	Robot Shadow Store	5	178	1 h 08 m	29 days	consistency	landed
INC-2025-131	2025-10-14	Map Authoring Studio	1	34	3 h 22 m	44 days	process bypass	partly landed

2. INC-2026-062 and the incident population examined - part 2 of 2

A further extract of the 2. INC-2026-062 and the incident population examined appears here, taken in order from the point at which the previous part ended. The sequence is unbroken either side of it.

Whatever caveat the 2. INC-2026-062 and the incident population examined carried at its head travels with these rows, and no additional caveat attaches to this part. Pagination alone determines where it falls.

Schedule F - Incident population examined, with follow-through status (TDR-2026-16) (continued, part 2 of 2)

Incident	Date	Component	Sites affected	Robots affected	Time to mitigate	Time to resolve	Root cause class	Remediation landed
INC-2025-139	2025-11-02	Event Bus	7	246	2 h 47 m	63 days	capacity	partly landed

Incident	Date	Component	Sites affected	Robots affected	Time to mitigate	Time to resolve	Root cause class	Remediation landed
INC-2025-146	2025-11-19	Billing Usage Meter	0	0	not applicable; caught before billing	11 days	process defect	landed
INC-2025-152	2025-12-04	Video Snippet Service	0	0	not applicable; retention only	37 days	data retention	partly landed
INC-2026-004	2026-01-13	Localization Engine	1	18	not a service outage; 5 days to a workaround	open at the review date	performance	not landed
INC-2026-009	2026-01-27	Gripper Controller	1	4	1 h 30 m to quarantine the affected robots	48 days	calibration	landed
INC-2026-015	2026-02-09	Config Distribution	2	7	44 m	22 days	validation	landed
INC-2026-021	2026-02-24	Simulation Harness	0	0	not applicable; developer tooling	open at the review date	tooling	not landed
INC-2026-028	2026-03-10	Fleet Health Monitor	1	22	not a service outage; 2 days of delayed repair	31 days	observability	landed
INC-2026-033	2026-03-25	Speed and Separation Monitor	1	6	immediate speed reduction; 3 h to review the fleet	open at the review date	safety margin	not landed
INC-2026-039	2026-04-07	Order Broker	1	1	1 h 04 m	19 days	concurrency	landed
INC-2026-045	2026-04-21	Edge Gateway Agent	1	39	5 h 12 m	26 days	observability	partly landed
INC-2026-051	2026-05-06	Site Provisioning Service	1	0	2 h 18 m	9 days	configuration drift	partly landed
INC-2026-057	2026-05-19	Battery Management Service	2	5	not a service outage; robots derated	open at the review date	modelling	not landed
INC-2026-062	2026-06-02	Release Orchestrator	0	0	not applicable; process only	open at the review date	process control	not landed
INC-2026-068	2026-06-16	Wheel Odometry Filter	1	14	not a service outage; 3 h of degraded operation	open at the review date	environmental	not landed

3. Detection paths and time to mitigate

Duration for incident INC-2025-090 splits into immediate stop, 22 m to return to service of impact and 41 days to close the underlying cause. Hardware fault event INC-2025-090 touched one site and 3 robots, and Fathom Mobility was the affected account. Remediation for Fathom Mobility incident INC-2025-090 is to add the connector torque check to incoming inspection and retrofit the affected harness batch; Bettina Vogel owns it and the committed date is 2025-09-30. The reviewer's reading of INC-2025-090 is that the team's analysis was honest about a gap it could have concealed. On 2025-07-19 the record for Fathom Mobility incident INC-2025-090 opens: a protective stop was triggered by a false positive on one safety channel while the second channel read clear. hardware fault incident INC-2025-090 has no single accountable owner, and that is the finding the reviewer would press hardest.

Cause analysis in Safety Controller incident INC-2025-090 concluded that a marginal connector on one channel produced intermittent readings inside the accepted noise band. Reported detection applies to Safety Controller incident INC-2025-090, since the robot self-reported the fault and the safety case review picked it up. Hardware fault incident

INC-2025-090 closure status is the inspection change landed on 2025-08-04; nine units await retrofit. Fathom Mobility incident INC-2025-090 would have been shorter or smaller had it not been that the connector torque specification was not part of the incoming inspection sheet. Safety Controller accounts for incident INC-2025-090, a hardware fault event. hardware fault event INC-2025-090 is the reason one section of TDR-2026-16 concludes narrowly rather than generally.

Limited; the log retention window had closed before the reviewer asked for the raw data is the evidence grade for Billing Usage Meter incident INC-2025-146, and the reader should not read severity into it. Larkspur Home incident INC-2025-146 is recorded as follows on delivery: landed on 2025-11-30. The reviewer treats one contributing factor in incident INC-2025-146 as the more important lesson: the runbook described the reset in prose rather than making it part of the command. Underneath incident INC-2025-146 lies the design decision that the re-run procedure required a manual ledger reset that the operator skipped. Taken with the rest of the record, INC-2025-146 indicates that management's severity rating is defensible and its remediation priority is not. Billing Usage Meter incident INC-2025-146 is the clearest instance in TDR-2026-16 of a control that exists on paper and is weaker in practice.

Reading INC-2025-146 from its first entry, on 2025-11-19 a re-run of a failed metering window double-counted usage for one customer. Remediation for incident INC-2025-146 is to fold the reset into the re-run command and make double-counting detectable in the ledger; Nadia Semenova owns it and the committed date is 2025-12-31. So far as reach is concerned, incident INC-2025-146 involved no production site and no robot directly, the exposure being to internal systems. Billing Usage Meter owns Larkspur Home incident INC-2025-146, which the reviewer classes as process defect. Mitigation of incident INC-2025-146 took not applicable; caught before billing and full resolution took 11 days. INC-2025-146 is the clearest instance in TDR-2026-16 of a control that exists on paper and is weaker in practice.

What went wrong in validation event INC-2026-015 was that precondition validation ran at publication time rather than at apply time. Taken with the rest of the record, Config Distribution incident INC-2026-015 indicates that the remediation addresses the immediate cause and leaves the contributing factor in place. Incident INC-2026-015 rests on moderate, resting on the team's own post-mortem and one supporting log extract evidence, which the reviewer states separately from the severity of the finding. In Fathom Mobility incident INC-2026-015 the first signal was the on-robot supervisor raised a validation fault on the next mission, and the reviewer treats that as reported detection. Validation incident INC-2026-015 began on 2026-02-09 when a robot that had been offline at publication applied a configuration bundle without re-validating preconditions. validation event INC-2026-015 is the only entry in its group where the reviewer and the engineering team reached different conclusions.

The commitment register shows validation incident INC-2026-015 against an undertaking to move precondition validation to apply time and add an offline apply case to the platform suite, with Sam Oyelaran named and 2026-03-31 committed. Validation event INC-2026-015 carries a second-order finding: the offline apply path was added for a customer with intermittent connectivity and was lightly reviewed. So far as reach is concerned, validation event INC-2026-015 involved 2 sites and 7 robots. 44 m elapsed before INC-2026-015 was mitigated, with resolution at 22 days. On delivery, validation event INC-2026-015 shows this: landed on 2026-03-03. validation incident INC-2026-015 is reported at the evidence strength available and has not been promoted to fit the narrative.

The immediate event in INC-2026-033 was that a speed reduction engaged later than expected when an operator approached faster than the assumed walking speed, dated 2026-03-25. Classifying safety margin event INC-2026-033 as safety margin matters because the Speed and Separation Monitor backlog contains two further items of the same kind. The reviewer checked the safety margin event INC-2026-033 commitment against the code and release records; not landed; a design study is running and the vendor assessment is booked. What Asterion Logistics incident INC-2026-033 tells the reviewer is that the commitment date was met and the verifying test now exists, which is not the pattern across the whole register. The corrective action for Speed and Separation Monitor incident INC-2026-033 is to measure closing rate where the sensing permits and re-run the protective distance calculation, and 2026-08-31 is the date the team gave. INC-2026-033 was cross-checked against the engineering record supplied for TDR-2026-16, and the two agree.

So far as reach is concerned, safety margin event INC-2026-033 involved one site and 6 robots. Speed and Separation Monitor incident INC-2026-033 would have been shorter or smaller had it not been that the assumption was documented in the internal safety case and accepted at the time. Reported detection applies to Asterion Logistics incident INC-2026-033, since the on-robot safety log flagged a margin breach that did not become a contact event. INC-2026-033 was caused by a design gap: the protective distance uses a fixed assumed approach speed rather than a measured closing

rate. Incident INC-2026-033 rests on moderate, resting on the team's own post-mortem and one supporting log extract evidence, which the reviewer states separately from the severity of the finding. INC-2026-033 was cross-checked against the engineering record supplied for TDR-2026-16, and the two agree.

INC-2026-039 was contained to one site and 1 robots. The reviewer's reading of incident INC-2026-039 is that an operational workaround masked the defect for longer than the incident record suggests. The two durations recorded for Crestline Fulfillment incident INC-2026-039 are 1 h 04 m to mitigate and 19 days to resolve. Attention came to concurrency incident INC-2026-039 when the site supervisor reported a blocked location. INC-2026-039 records that on 2026-04-07, an order cancellation raced with assignment and left an orphaned task that blocked a pick face for an hour. concurrency incident INC-2026-039 illustrates why TDR-2026-16 separates the severity of a finding from the confidence behind it.

Verification of the concurrency incident INC-2026-039 remediation gives landed on 2026-04-26. A contributing factor in Crestline Fulfillment incident INC-2026-039 was that the journal reconciliation runs nightly, so the orphan was invisible until the morning. The classification applied to Order Broker incident INC-2026-039 is concurrency, and the affected component is Order Broker. Incident INC-2026-039 produced one commitment: to serialise cancellation against assignment and add an hourly orphan sweep, owned by Grace Lindqvist, dated 2026-05-31. Order Broker incident INC-2026-039 carries strong, resting on the production traces the team supplied evidence, and the reviewer declined to raise the finding beyond what that supports. concurrency event INC-2026-039 was the subject of a specific management representation, which the reviewer has recorded but not relied on alone.

INC-2025-063 was caused by a design gap: the allocator filter excluded low-charge robots without checking whether a charge slot was actually available. Contributing to Granite Peak Distribution incident INC-2025-063, and not repaired by the remediation, is that the charge scheduler and the allocator hold separate views of charger availability. The commitment register shows policy defect incident INC-2025-063 against an undertaking to make the allocator consult charger availability before excluding a robot, with Marta Kellerman named and 2025-07-31 committed. Assessed on its own record, incident INC-2025-063 shows that the commitment date was met and the verifying test now exists, which is not the pattern across the whole register. How incident INC-2025-063 surfaced matters to the assessment: an operator noticed idle robots and raised a support ticket. Granite Peak Distribution incident INC-2025-063 is one of the entries where TDR-2026-16 relies on the company's own measurement rather than an independent one.

Incident INC-2025-063 belongs to the policy defect group and sits with Fleet Allocator. INC-2025-063 took 6 h 05 m to stabilise and 12 days to close out. Fleet Allocator incident INC-2025-063 is recorded as follows on delivery: landed on 2025-06-08. INC-2025-063 affected 3 sites and 28 robots. On 2025-05-27 the record for policy defect event INC-2025-063 opens: robots parked below the battery reserve threshold were never offered work and the site ran short of movers. INC-2025-063 was tested rather than accepted, and the test is described in the working papers behind TDR-2026-16.

Underneath INC-2026-051 lies the design decision that the site template had been forked for a customer network requirement and drifted from the baseline. The classification applied to Site Provisioning Service incident INC-2026-051 is configuration drift, and the affected component is Site Provisioning Service. After incident INC-2026-051, Marcus Duplessis took the action to check every template against the baseline and re-converge the four forked templates by 2026-06-15. Evidence strength on the INC-2026-051 analysis is strong, resting on the production traces the team supplied. Site Provisioning Service incident INC-2026-051 carries a second-order finding: template drift was not detected because the baseline check compares only the default template. configuration drift event INC-2026-051 has an owner who accepted the finding immediately, which is worth recording alongside the finding itself.

Attention came to configuration drift incident INC-2026-051 when the identity service rejected the site's first token request. 2 h 18 m elapsed before Larkspur Home incident INC-2026-051 was mitigated, with resolution at 9 days. Reading Site Provisioning Service incident INC-2026-051 from its first entry, on 2026-05-06 a new site went live with a bespoke network template that omitted the outbound certificate pin. Assessed on its own record, configuration drift event INC-2026-051 shows that the same class of defect can recur in two sibling components that were not changed. On delivery, configuration drift event INC-2026-051 shows this: the check landed on 2026-05-15; two templates remain forked. Site Provisioning Service incident INC-2026-051 was cross-checked against the engineering record supplied for TDR-2026-16, and the two agree.

The immediate event in Wheel Odometry Filter incident INC-2026-068 was that odometry drift exceeded specification on a wet floor and localization confidence fell below the mission threshold, dated 2026-06-16. The reviewer checked the Wheel Odometry Filter incident INC-2026-068 commitment against the code and release records; not landed; the detector is prototyped and unvalidated. Harborline Systems incident INC-2026-068 carries a second-order finding: the site's cleaning schedule changed and now overlaps the peak shift. Harborline Systems felt Harborline Systems incident INC-2026-068, which affected one site and 14 robots. Wheel Odometry Filter incident INC-2026-068 was detected because robots stopped and requested assistance, which the site reported as unplanned downtime. Harborline Systems incident INC-2026-068 is one of the entries where TDR-2026-16 relies on the company's own measurement rather than an independent one.

Drawing the evidence in Wheel Odometry Filter incident INC-2026-068 together, the blast radius was set by a deployment topology decision made years earlier. The remediation committed after INC-2026-068 was to add a slip detector that flags the condition and issue a site advisory on cleaning windows, owned by Yuki Harada with a due date of 2026-09-15. Wheel Odometry Filter owns Harborline Systems incident INC-2026-068, which the reviewer classes as environmental. The two durations recorded for INC-2026-068 are not a service outage; 3 h of degraded operation to mitigate and open at the review date to resolve. Evidence strength on the Wheel Odometry Filter incident INC-2026-068 analysis is limited; the log retention window had closed before the reviewer asked for the raw data. INC-2026-068 was raised again with the owning engineer after the initial interview, and the position did not change.

4. Root causes, contributing factors and recurrence risk

On 2025-10-14 the record for Map Authoring Studio incident INC-2025-131 opens: an unpublished map draft was exported and hand-loaded at a site and robots ran against unreviewed geometry. Process bypass incident INC-2025-131 shows a short mitigation and a long tail: 3 h 22 m against 44 days. The blast radius of process bypass incident INC-2025-131 was one site and 34 robots. Evidence strength on the INC-2025-131 analysis is strong, resting on the production traces the team supplied. On delivery, process bypass event INC-2025-131 shows this: the export restriction landed on 2025-11-27; the publication work is open. incident INC-2025-131 is one of the entries where TDR-2026-16 relies on the company's own measurement rather than an independent one.

Contributing to incident INC-2025-131, and not repaired by the remediation, is that the site had a standing workaround for slow map publication that predated the review step. Incident INC-2025-131 supports one conclusion and not another: the failure was in the detection design rather than in the code that failed. Cause analysis in incident INC-2025-131 concluded that the export function did not check publication state. Process bypass event INC-2025-131 belongs to the process bypass group and sits with Map Authoring Studio. Lucia Ferrante committed after Juniper Bay Apparel incident INC-2025-131 to restrict export to published revisions and shorten the publication path so the workaround is unnecessary, due 2025-12-15. incident INC-2025-131 appears twice in the working papers for TDR-2026-16 because the first answer was corrected after a follow-up request.

Data retention incident INC-2025-152 touched no production site and no robot directly, the exposure being to internal systems, and internal was the affected account. Anjali Sekhon committed after data retention event INC-2025-152 to apply the lifecycle rule through the configuration baseline and delete the over-retained clips, due 2026-01-31. Data retention event INC-2025-152 belongs to the data retention group and sits with Video Snippet Service. On 2025-12-04 the record for data retention event INC-2025-152 opens: one region retained operator video clips for ninety days against a thirty-day commitment. Root cause in data retention incident INC-2025-152 was that the lifecycle rule was applied per storage bucket and one region's bucket was created later without it. Video Snippet Service incident INC-2025-152 is treated as material to the investment case rather than as a routine engineering item.

For data retention incident INC-2025-152 the reviewer holds moderate, resting on the team's own post-mortem and one supporting log extract evidence and reports the finding at that strength. As to whether the Video Snippet Service incident INC-2025-152 remediation actually landed: the rule landed on 2026-01-10; deletion of the historical clips is open. Taken with the rest of the record, INC-2025-152 indicates that the incident is unremarkable in itself and revealing about the observability of the surrounding system. The contributing factor recorded against internal incident INC-2025-152 is that bucket creation was not covered by the configuration baseline check. Time to mitigate data retention event

INC-2025-152 was not applicable; retention only, and time to resolve was 37 days. incident INC-2025-152 is recorded here without adjustment; the reviewer saw no basis to restate what the company reported.

The reviewer grades the Dunbar Industrial incident INC-2025-058 evidence as moderate, resting on the team's own post-mortem and one supporting log extract and has not upgraded the finding to compensate. Landed on 2025-05-30; the backfill was completed on 2025-06-03 is the position on the data integrity event INC-2025-058 commitment as at the review date. The corrective action for data integrity incident INC-2025-058 is to key suppression on the tenant and identifier pair and backfill the affected order records, and 2025-07-15 is the date the team gave. Data integrity event INC-2025-058 affected 2 site or tenant with no robot taken out of service. Data integrity is the class of incident INC-2025-058; WMS Connector is the component. WMS Connector incident INC-2025-058 is not material on its own and belongs in this report because of what it reveals about the surrounding process.

Detection came through the customer reported missing picks the same afternoon, which places data integrity incident INC-2025-058 on the reported side of the ledger. The immediate event in Dunbar Industrial incident INC-2025-058 was that two tenants reused the same order identifier namespace and duplicate suppression discarded live orders, dated 2025-05-09. INC-2025-058 is not read as an isolated failure: the commitment date was met and the verifying test now exists, which is not the pattern across the whole register. A contributing factor in Dunbar Industrial incident INC-2025-058 was that the integration specification allowed the identifier to be tenant-scoped but did not require it. The engineering team's root cause for Dunbar Industrial incident INC-2025-058 is that duplicate suppression keyed on a customer-supplied identifier alone rather than on tenant and identifier, and the reviewer found the analysis sound. WMS Connector incident INC-2025-058 illustrates why TDR-2026-16 separates the severity of a finding from the confidence behind it.

In Evergreen Warehousing incident INC-2025-052 the first signal was the site's own shift report, escalated the next morning, and the reviewer treats that as reported detection. Scheduling incident INC-2025-052 supports one conclusion and not another: the failure was in the detection design rather than in the code that failed. On scheduling event INC-2025-052 the evidence is moderate; the narrative is consistent but the reviewer could not independently reproduce it; the severity assessment is stated separately below. In scheduling event INC-2025-052, the site had changed its shift pattern without notifying the deployment team, which is a process finding rather than a code finding. Charge Scheduler accounts for scheduling incident INC-2025-052, a scheduling event. scheduling incident INC-2025-052 was compared with the equivalent position at two other components, and it is neither the best nor the worst.

Duration for scheduling event INC-2025-052 splits into 4 h 10 m of impact and 16 days to close the underlying cause. Scheduling event INC-2025-052 began on 2025-04-21 when an early shift end left opportunity-charge windows reserved and the following shift started with a depleted fleet. The affected population in scheduling event INC-2025-052 was one site and 44 robots. The remediation committed after Evergreen Warehousing incident INC-2025-052 was to release charge windows on a shift-end event and reconcile the calendar nightly, owned by Nadia Semenova with a due date of 2025-06-30. Underneath INC-2025-052 lies the design decision that the scheduler treated a shift calendar entry as immutable once the window had opened. scheduling incident INC-2025-052 was reviewed against the position twelve months earlier, and the direction of travel is favourable.

The post-mortem for INC-2025-110 identifies the cause as follows: the loader treated a missing partition as empty rather than as incomplete. Farida Cisse committed after data integrity incident INC-2025-110 to fail the load on an incomplete partition and reprocess the affected days, due 2025-10-31. For incident INC-2025-110 the reviewer holds limited; the log retention window had closed before the reviewer asked for the raw data evidence and reports the finding at that strength. Analytics Warehouse Loader incident INC-2025-110 was raised on 2025-09-03 after a late telemetry partition was excluded from the daily load and a customer report understated throughput. Attention came to Analytics Warehouse Loader incident INC-2025-110 when the customer challenged the report three days later. INC-2025-110 was verified to the extent the available evidence permits, and the residual uncertainty is stated rather than absorbed.

Drawing the evidence in data integrity incident INC-2025-110 together, the fix is real and the regression protection is not, so recurrence remains possible. Beyond the direct cause, Analytics Warehouse Loader incident INC-2025-110 shows that the partition completeness check was written but never wired into the load gate. In taxonomy terms data integrity incident INC-2025-110 is a data integrity incident in Analytics Warehouse Loader. Landed on 2025-09-21 with a reprocessing run for the affected week is the position on the Analytics Warehouse Loader incident INC-2025-110

commitment as at the review date. INC-2025-110 touched 1 site or tenant with no robot taken out of service, and Kestrel Medical Supply was the affected account. INC-2025-110 would move from a concern to a verified defect if one further measurement were supplied.

Classifying availability incident INC-2025-071 as availability matters because the Telemetry Ingest backlog contains two further items of the same kind. Incident INC-2025-071 traces back to a single fact: a broker partition lost its leader during a rolling upgrade of the streaming cluster. 52 m elapsed before INC-2025-071 was mitigated, with resolution at 52 m. In Telemetry Ingest incident INC-2025-071 the first signal was the ingest availability alert fired within ninety seconds, and the reviewer treats that as alerted detection. Evidence strength on the INC-2025-071 analysis is moderate; the narrative is consistent but the reviewer could not independently reproduce it. incident INC-2025-071 is reported at the evidence strength available and has not been promoted to fit the narrative.

Drawing the evidence in availability event INC-2025-071 together, an operational workaround masked the defect for longer than the incident record suggests. Remediation for INC-2025-071 is to add a leadership and under-replicated partition check to the broker upgrade runbook; Erik Palmgren owns it and the committed date is 2025-07-31. In multiple sites incident INC-2025-071, the upgrade runbook did not require a partition leadership check between broker restarts, which is a process finding rather than a code finding. The affected population in multiple sites incident INC-2025-071 was 6 sites and 214 robots. Multiple sites incident INC-2025-071 is recorded as follows on delivery: landed on 2025-07-18 and exercised in the September failover drill. availability event INC-2025-071 was reviewed against the position twelve months earlier, and the direction of travel is favourable.

Schedule G - Open defect population and ageing by service (TDR-2026-16)

Service	Open defects	Oldest open item	Severity mix	Closed in reviewed period	Engineering owner
Motion Planner	14	2025-04-06	0 high, 1 medium, 13 low	7	Priya Raghunathan
Costmap Fusion	21	2025-07-22	0 high, 4 medium, 17 low	9	Daniel Okafor
Fleet Allocator	9	2025-06-27	0 high, 3 medium, 6 low	6	Marta Kellerman
Traffic Manager	12	2025-03-27	2 high, 2 medium, 8 low	2	Owen Brantley
Charge Scheduler	7	2025-09-29	2 high, 3 medium, 2 low	10	Nadia Semenova
Safety Controller	3	2025-04-20	1 high, 2 medium, 0 low	4	Bettina Vogel
E-Stop Chain	2	2025-12-01	2 high, 2 medium, 0 low	6	Theo Bassett
Lidar Driver	18	2025-08-03	1 high, 5 medium, 12 low	7	Yuki Harada
Localization Engine	16	2025-11-30	1 high, 4 medium, 11 low	7	Rahul Menon
WMS Connector	23	2025-11-30	0 high, 10 medium, 13 low	11	Tomas Villareal
Order Broker	11	2025-09-11	0 high, 4 medium, 7 low	4	Grace Lindqvist
Telemetry Ingest	19	2025-05-12	0 high, 9 medium, 10 low	6	Farida Cisse
OTA Updater	8	2026-02-14	1 high, 3 medium, 4 low	1	Erik Palmgren
Identity and Authorization	6	2026-04-25	1 high, 1 medium, 4 low	3	Sonia Krall
Remote Operations Console	27	2025-12-01	0 high, 5 medium, 22 low	8	Anjali Sekhon
Simulation Harness	31	2025-11-19	1 high, 14 medium, 16 low	11	Colin Achebe
Hardware-in-the-Loop Rig	13	2026-01-03	0 high, 2 medium, 11 low	6	Nathan Oduya

4. Root causes, contributing factors and recurrence risk - part 2 of 2

Following on directly, these further rows of the 4. Root causes, contributing factors and recurrence risk belong to the same schedule and share its numbering. The break is editorial.

Treat the conventions declared for the 4. Root causes, contributing factors and recurrence risk as governing here too; this part introduces none of its own. No row is omitted at the join.

Schedule G - Open defect population and ageing by service (TDR-2026-16) (continued, part 2 of 2)

Service	Open defects	Oldest open item	Severity mix	Closed in reviewed period	Engineering owner
Fleet Health Monitor	17	2026-01-29	1 high, 8 medium, 8 low	14	Wanda Njoku
Battery Management Service	10	2025-03-22	2 high, 1 medium, 7 low	3	Peter Halvorsen
Gripper Controller	15	2025-08-13	1 high, 1 medium, 13 low	11	Ines Duarte
Map Authoring Studio	22	2025-04-08	1 high, 6 medium, 15 low	3	Lucia Ferrante
Site Provisioning Service	5	2025-12-13	0 high, 2 medium, 3 low	3	Marcus Duplessis
Event Bus	4	2025-06-20	1 high, 1 medium, 2 low	5	Erik Palmgren
Config Distribution	6	2026-04-19	2 high, 1 medium, 3 low	3	Sam Oyelaran
Robot Shadow Store	9	2025-04-12	1 high, 2 medium, 6 low	8	Marta Kellerman
Analytics Warehouse Loader	20	2026-03-03	0 high, 3 medium, 17 low	14	Farida Cisse
Billing Usage Meter	7	2025-09-17	1 high, 2 medium, 4 low	4	Nadia Semenova
Alarm Router	14	2026-01-16	1 high, 3 medium, 10 low	4	Wanda Njoku
Video Snippet Service	12	2025-09-24	2 high, 5 medium, 5 low	1	Anjali Sekhon
Speed and Separation Monitor	4	2026-04-27	2 high, 1 medium, 1 low	3	Bettina Vogel
Wheel Odometry Filter	8	2025-07-27	2 high, 3 medium, 3 low	12	Yuki Harada
Payload Scale Interface	11	2025-05-18	0 high, 2 medium, 9 low	5	Ines Duarte
Edge Gateway Agent	5	2025-04-14	0 high, 2 medium, 3 low	1	Sam Oyelaran
Release Orchestrator	16	2026-04-20	2 high, 5 medium, 9 low	7	Colin Achebe

5. Reliability of the services carrying the incidents

Hosting for Payload Scale Interface is on-robot Linux, user-space driver, and its C++17 code writes through to a local calibration record. 11 defects remain open against Payload Scale Interface; the one the reviewer pressed on is that the calibration record is stored locally and is lost when the compute module is swapped. For Payload Scale Interface the stated target is that payload mass is reported within 2 per cent of the reference load, expressed against production traffic rather than a benchmark. Automated test coverage on Payload Scale Interface stands at 36 per cent line coverage. Ines Duarte leads Robot Firmware's work on Payload Scale Interface and was the reviewer's principal interview for it. Payload Scale Interface is included because a reader of TDR-2026-16 should be able to disagree with the inference without repeating the work.

For Payload Scale Interface the reviewer records that there is no finding of unfitness; there is a finding that the team cannot currently demonstrate fitness. Supportability of Payload Scale Interface turns on one fact: none identified. The dependency map places the load cell hardware above Payload Scale Interface and the gripper controller and the order execution path below it. Evidence examined for Payload Scale Interface comprised the interface specification, calibration records for nine units and a module-swap test. Placement of Payload Scale Interface is edge, only on robots fitted with the weighing deck, which the reviewer tested by tracing a request end to end. Payload Scale Interface would be a candidate for the first post-close engineering review under any reasonable plan.

Documents and data inspected for OTA Updater were the update state machine specification, signing key custody notes and two rollback drills. For OTA Updater the reviewer records that the reviewer accepts the engineering position while noting that the verifying test does not exist. OTA Updater sits at 76 per cent line coverage; the trend across the reviewed period was down by two points after a large refactor. OTA Updater carries 8 open defects, among them a case

where a paused ring resumes automatically after a service restart, which surprised two site teams. The cadence recorded for OTA Updater is every three weeks, following the autonomy release, which matches the observed release history. OTA Updater was raised again with the owning engineer after the initial interview, and the position did not change.

OTA Updater has a single named owner, Erik Palmgren of Platform Infrastructure, which is not true of every service reviewed. The ageing position for OTA Updater is none; artefact signing was moved to the current key format in 2025, and the reviewer checked it against the vendor statements on file. Interface discipline around OTA Updater is adequate; the schema is versioned and change is gated on a compatibility check. Implementation of OTA Updater rests on Rust 1.79 on the robot, Go 1.22 in the cloud; the process runs on split: an on-robot agent and a cloud release service and persists to content-addressed object storage for artefacts, PostgreSQL 14 for state. For OTA Updater the stated target is that an update is either fully applied or fully rolled back, with no mixed-version robot, expressed against production traffic rather than a benchmark. OTA Updater is reported at the evidence strength available and has not been promoted to fit the narrative.

Event Bus commits that publish acknowledgement inside 25 ms at p95 and no partition without a leader, and the objective is measured continuously rather than sampled. Erik Palmgren of Platform Infrastructure maintains Event Bus, and the service inventory entry was current at the review date. Event Bus sits at no meaningful automated unit coverage, verification being by witnessed test instead; the trend across the reviewed period was up by seven points. The manifests place Event Bus cloud, one cluster per region, three availability zones. Event Bus was built in operated open-source broker, no in-house code and has not been rewritten; it still runs on cloud, a managed streaming cluster against the broker's own log storage with seven-day retention. Event Bus is the only entry in its group where the reviewer and the engineering team reached different conclusions.

Event Bus carries 4 open defects, among them a case where one topic carries both control and telemetry traffic, so a telemetry backlog delays control events. The ageing position for Event Bus is the broker minor release in production leaves the vendor's support window in 2027, and the reviewer checked it against the vendor statements on file. The cadence recorded for Event Bus is upgraded twice a year, which matches the observed release history. Every publishing service feed Event Bus; a failure in Event Bus is felt by every consuming service. On Event Bus the conclusion is qualified: the risk is concentrated in one code path that has neither automated coverage nor a bench equivalent. Event Bus was verified to the extent the available evidence permits, and the residual uncertainty is stated rather than absorbed.

Inputs to Remote Operations Console come from the identity service, the telemetry ingest path and the video snippet service, and its consumers are the traffic manager and the alarm router. For Remote Operations Console the stated target is that operator input reaches the robot inside 350 ms at p95 on a supported network, expressed against production traffic rather than a benchmark. The assessment of Remote Operations Console is not unqualified: no verified defect was identified, and two plausible concerns are recorded below at low evidence strength. Implementation of Remote Operations Console rests on TypeScript with a React front end; the process runs on cloud, a static bundle plus a session gateway and persists to PostgreSQL 14 for session records and object storage for clips. The cadence recorded for Remote Operations Console is twice a week, which matches the observed release history. Remote Operations Console would be a candidate for the first post-close engineering review under any reasonable plan.

Interface discipline around Remote Operations Console is adequate; the schema is versioned and change is gated on a compatibility check. For Remote Operations Console the supportability position is a charting library that no longer receives security updates. The evidence base for Remote Operations Console was the console interaction specification, session latency measurements and an accessibility walk-through. Responsibility for Remote Operations Console was confirmed with Anjali Sekhon, whose team, Applications, also holds the on-call duty. Remote Operations Console sits at 52 per cent line coverage; the trend across the reviewed period was flat, with a one-off drop when a test suite was quarantined. Remote Operations Console was discussed with two engineers who gave slightly different accounts, and the difference is preserved here.

Evidence examined for WMS Connector comprised six tenant integration specifications and a month of connector error logs. Behind WMS Connector sits PostgreSQL 14 plus a message outbox table; the service itself is Java 21 on cloud, containers on a managed Kubernetes cluster. The objective recorded against WMS Connector is that order acknowledgement returned to the warehouse system inside 1.5 s at p95, and it is reported weekly to the engineering leadership. Taking the evidence together, WMS Connector the design is sound and the test evidence is thinner than the

design deserves. The topology for WMS Connector is cloud, one deployment per customer tenant, and the reviewer confirmed it against the deployment manifests. WMS Connector is recorded here without adjustment; the reviewer saw no basis to restate what the company reported.

WMS Connector sits between the customer warehouse management systems and the identity service and the order broker and the billing usage meter, and that position is what makes its availability objective load-bearing. 23 defects remain open against WMS Connector; the one the reviewer pressed on is that duplicate order suppression relies on a customer-supplied identifier that two tenants reuse. WMS Connector publishes a stable interface, though two fields used by the order broker and the billing usage meter are documented only in code comments. On paper WMS Connector releases every two weeks, per tenant on request, and the observed practice did not diverge from that. Against WMS Connector, the coverage figure is 69 per cent line coverage, which the reviewer regards that as adequate for the risk this component carries. WMS Connector is treated as material to the investment case rather than as a routine engineering item.

Telemetry Ingest was checked for shared-database coupling, and the ownership boundary held. Read against its objective, Telemetry Ingest the risk is concentrated in one code path that has neither automated coverage nor a bench equivalent. In production Telemetry Ingest is cloud, one ingest endpoint per region with an edge buffer on the robot, unchanged since the last architecture review. Responsibility for Telemetry Ingest was confirmed with Farida Cisse, whose team, Data and Telemetry, also holds the on-call duty. Telemetry Ingest releases weekly, with schema changes gated on a compatibility check. Telemetry Ingest was reviewed against the position twelve months earlier, and the direction of travel is favourable.

End-of-life review of Telemetry Ingest found the stream processor runs a release that leaves community support in late 2026. What the reviewer had for Telemetry Ingest was the telemetry schema registry, retention configuration and a forced-offline buffer test; the conclusions below do not reach past it. The edge gateway agent and the event bus feed Telemetry Ingest; a failure in Telemetry Ingest is felt by the fleet health monitor, the analytics warehouse loader and the alarm router. Telemetry Ingest is written in Go 1.22 with a Rust decoder, executes on cloud, containers plus a stream processor, and keeps its state in a time-series store with ninety-day retention and object storage beyond that. Telemetry Ingest sits at 61 per cent line coverage; the trend across the reviewed period was up by three points. Telemetry Ingest was checked against the corresponding entry in the company's own register, and the register was current.

Video Snippet Service was assessed on the retention configuration, a lifecycle audit across regions and the access log sample, and nothing beyond that was available in the data room. Video Snippet Service has a single named owner, Anjali Sekhon of Applications, which is not true of every service reviewed. On end-of-life exposure, Video Snippet Service shows a transcoder build with known but low-severity upstream advisories. Video Snippet Service sits at 43 per cent line coverage; the trend across the reviewed period was up by four points. The reviewer's judgement on Video Snippet Service is that the reviewer could corroborate the availability claim but not the latency claim, which rests on a single dashboard. Video Snippet Service is reported here in full because a summary of it in an earlier draft of TDR-2026-16 was misread as an assurance.

Upstream of Video Snippet Service sit the on-robot camera buffer and the identity service; downstream, the remote operations console and the incident review workflow consume its output. The topology for Video Snippet Service is cloud, regional, with an on-robot capture buffer, and the reviewer confirmed it against the deployment manifests. Video Snippet Service pairs a Go 1.22 with a media pipeline implementation with cloud, containers plus a media transcoder, and its data lives in object storage with a thirty-day lifecycle and a PostgreSQL 14 index. For Video Snippet Service the stated target is that a requested snippet is available to the operator inside 45 s, expressed against production traffic rather than a benchmark. The cadence recorded for Video Snippet Service is monthly, which matches the observed release history. Video Snippet Service was inspected in the repository rather than described from the ticket, which is why the account is specific.

Schedule H - Change volume, change failure rate and restore time by month (TDR-2026-16)

Month	Changes to production	Failed changes	Change failure rate	Mean time to restore	Rollbacks
2025-01	68	6	8.8%	39 min	1
2025-02	40	0	0.0%	22 min	2
2025-03	65	2	3.0%	64 min	0
2025-04	75	3	4.0%	48 min	0

Month	Changes to production	Failed changes	Change failure rate	Mean time to restore	Rollbacks
2025-05	44	3	6.8%	62 min	2
2025-06	42	0	0.0%	46 min	2
2025-07	75	1	1.3%	59 min	1
2025-08	39	3	7.6%	32 min	3
2025-09	49	4	8.1%	41 min	1
2025-10	71	5	7.0%	41 min	3
2025-11	77	7	9.0%	85 min	1
2025-12	80	1	1.2%	53 min	0
2026-01	51	4	7.8%	43 min	1
2026-02	54	5	9.2%	59 min	2
2026-03	62	1	1.6%	37 min	3
2026-04	50	2	4.0%	78 min	2
2026-05	78	3	3.8%	46 min	1
2026-06	43	1	2.3%	34 min	0
2026-07	55	4	7.2%	30 min	0
2026-08	52	2	3.8%	19 min	3

6. Where the incidents landed: sites and customers

At Meridian, Idaho the fleet is 96 robots, and the site has been live since 2022-09. Remaining capacity at Meridian, Idaho is 22%, measured against the site's peak order profile. Meridian, Idaho has an owner who accepted the finding immediately, which is worth recording alongside the finding itself.

Against its peak profile, Fremont, California retains not measured headroom on a test harness integration. Fremont, California is the reference site, and the only one with a laboratory alongside, which is why the reviewer sampled it. Fremont, California is one of the items the reviewer would put on a monitoring pack rather than on a closing checklist.

On Grand Prairie, Texas the record shows that it is the second Asterion site, commissioned on the same template. Headroom at Grand Prairie, Texas stands at 31%; vendor A is the integrated warehouse system. Grand Prairie, Texas was the subject of a specific management representation, which the reviewer has recorded but not relied on alone.

37 robots are deployed at Bethlehem, Pennsylvania, which went live in 2025-11. 51% of headroom remains at Bethlehem, Pennsylvania, whose connector targets vendor A. Bethlehem, Pennsylvania is the only entry in its group where the reviewer and the engineering team reached different conclusions.

Edwardsville, Illinois was included in the sample because it is the newest Crestline site, still ramping. Edwardsville, Illinois runs 24 robots for Crestline Fulfillment and opened in 2026-03. Edwardsville, Illinois is the entry that changed most between the first and second document requests behind TDR-2026-16.

Remaining capacity at Lehigh Valley, Pennsylvania is 38%, measured against the site's peak order profile. On Lehigh Valley, Pennsylvania the record shows that it is commissioned during the connector rework. Lehigh Valley, Pennsylvania was cross-checked against the engineering record supplied for TDR-2026-16, and the two agree.

The reason Chino, California appears here is that it is the go-live delayed by the forked provisioning template. 33 robots are deployed at Chino, California, which went live in 2026-01. Chino, California is one of the entries where TDR-2026-16 relies on the company's own measurement rather than an independent one.

2024-11 is the go-live month at Kenosha, Wisconsin, where Dunbar Industrial now runs 41 robots. For Kenosha, Wisconsin the material fact is that it is the second opportunity-charging site. Kenosha, Wisconsin has an owner who accepted the finding immediately, which is worth recording alongside the finding itself.

For Whitestown, Indiana the material fact is that it is the site with the standing map publication workaround. Remaining capacity at Whitestown, Indiana is 45%, measured against the site's peak order profile. Whitestown, Indiana was raised again with the owning engineer after the initial interview, and the position did not change.

For Reno, Nevada the material fact is that it is the smallest customer deployment. Vendor D is the warehouse system at Reno, Nevada, where headroom is 57%. Reno, Nevada has no single accountable owner, and that is the finding the reviewer would press hardest.

Schedule J - Site deployments, capacity headroom and integration target (TDR-2026-16)

Site	Customer	Robots deployed	Go-live	Capacity headroom	Warehouse system	Note

7. Security findings bearing on availability

Assessment finding SEC-2026-11 rests on the observation that file permissions observed on a laboratory robot; the reviewer holds that to be strong. Assessment finding SEC-2026-11 is not fully retired by its remediation: medium; the safety controller limit is the compensating control. SEC-2026-11 is assessed as follows: management's acceptance is defensible on the evidence and should be revisited before the fleet doubles. Finding SEC-2026-11 was opened on 2026-04-19 and states that gripper controller parameter store writable by any on-robot process. finding SEC-2026-11 is the clearest instance in TDR-2026-16 of a control that exists on paper and is weaker in practice.

Moderate, because the demonstration was performed in a laboratory tenant rather than in production evidence supports SEC-2026-08, namely that role grant performed in a staging tenant. Assessment finding SEC-2026-08 was opened on 2026-04-18 and states that identity service admin role assignable without a second approval. Finding SEC-2026-08 stands remediated; 2026-06-30 is what the record gives for closure. Medium-severity finding SEC-2026-08 translates into the following exposure: one administrator can grant themselves tenant-wide access. assessment finding SEC-2026-08 is included because a reader of TDR-2026-16 should be able to disagree with the inference without repeating the work.

What remains after assessment finding SEC-2026-09 is remediated is low. Finding SEC-2026-09 stands remediated; 2026-05-08 is what the record gives for closure. The finding at Fleet Health Monitor finding SEC-2026-09, dated 2026-04-18, is that fleet health monitor query interface allowed unbounded result sets. At 3.9 the score on assessment finding SEC-2026-09 places it in the Low group. assessment finding SEC-2026-09 would move from a concern to a verified defect if one further measurement were supplied.

Remediated is the status of Identity and Authorization finding SEC-2025-01, dated 2025-11-14. The residual position on Identity and Authorization finding SEC-2025-01 is low once lifetimes are cut to one hour with refresh, stated after mitigation rather than assuming full success. The reviewer declines to soften High-severity finding SEC-2025-01: the reviewer would not treat this as a closing condition and would put it on the first post-close sprint. For High-severity finding SEC-2025-01 the record shows that reproduced by the tester and confirmed in the token audit, which is moderate, because the demonstration was performed in a laboratory tenant rather than in production evidence and is graded separately from severity. SEC-2025-01 against Identity and Authorization carries a follow-up request that was satisfied after the draft of TDR-2026-16 was circulated.

The reason SEC-2026-10 against Billing Usage Meter was not downgraded is that an export could be altered after the fact. Strong evidence supports Billing Usage Meter finding SEC-2026-10, namely that bucket configuration read from the cloud export. After the fix, assessment finding SEC-2026-10 carries residual risk described as low; object lock is applied and verified nightly. Weighing SEC-2026-10 against Billing Usage Meter against the compensating controls, management's acceptance is defensible on the evidence and should be revisited before the fleet doubles. SEC-2026-10 was cross-checked against the engineering record supplied for TDR-2026-16, and the two agree.

Low-severity finding SEC-2025-12 is supported by the fact that permissions read from the warehouse grant inventory; evidence strength is moderate. Not planned is the closure position on finding SEC-2025-12, which the ledger records as accepted. Low-severity finding SEC-2025-12 was graded Low, band 4.2, by Redwater Security engagement RW-2025-3. The reviewer's position on assessment finding SEC-2025-12 is that the retest evidence is a screenshot rather than a re-run of the original test, which the reviewer notes. SEC-2025-12 against Analytics Warehouse Loader is not material on its own and belongs in this report because of what it reveals about the surrounding process.

Residual exposure on SEC-2025-04 reads medium; the shortened publication path is still outstanding. For SEC-2025-04 the record shows that reproduced, and independently observed in a field incident, which is limited, resting on a configuration read rather than on a demonstrated exploitation evidence and is graded separately from severity. 2025-11-27 is the closure position on assessment finding SEC-2025-04, which the ledger records as remediated. In finding SEC-2025-04 the exploitation path means an authenticated author can distribute unreviewed geometry. Map Authoring Studio finding SEC-2025-04 carries a follow-up request that was satisfied after the draft of TDR-2026-16 was circulated.

SEC-2025-08 against Event Bus is rated Medium with a numeric band of 6.0 on the tester's scoring scale. Event bus administrative interface reachable from the application subnet is the substance of Event Bus finding SEC-2025-08, raised on 2025-09-23. Read alongside the remediation record, SEC-2025-08 shows that the exposure is internal rather than

customer-facing, which lowers the severity without removing the finding. The remediation ledger shows Medium-severity finding SEC-2025-08 as remediated, 2025-10-31. SEC-2025-08 against Event Bus was reviewed against the position twelve months earlier, and the direction of travel is favourable.

Impact of SEC-2025-02 against Release Orchestrator is not theoretical; any engineer with build log access could replay a service credential. High-severity finding SEC-2025-02 is assessed as follows: the remediation is adequate and the time to remediate is not. What remains after assessment finding SEC-2025-02 is remediated is low; redaction is enforced in the logging library. The evidence for Release Orchestrator finding SEC-2025-02 is moderate: the tester extracted a working token from a retained log. finding SEC-2025-02 is recorded here without adjustment; the reviewer saw no basis to restate what the company reported.

For SEC-2026-17 against Motion Planner the reviewer notes that the retest evidence is a screenshot rather than a re-run of the original test, which the reviewer notes. 2026-06-19 is the closure position on SEC-2026-17, which the ledger records as remediated. SEC-2026-17 translates into the following exposure: a local compromise could alter planner limits without detection. Motion Planner finding SEC-2026-17 is rated High with a numeric band of 7.9 on the tester's scoring scale. finding SEC-2026-17 is one of the entries where TDR-2026-16 relies on the company's own measurement rather than an independent one.

The evidence for finding SEC-2025-03 is strong: demonstrated in the tester's laboratory network, not on a customer site. Assessment finding SEC-2025-03 was opened on 2025-09-18 and states that remote operations session relay accepted an unpinned certificate chain. Residual risk after remediation of SEC-2025-03 is low; pinning is applied at the gateway and the agent. Assessment finding SEC-2025-03 is assessed as follows: the retest evidence is a screenshot rather than a re-run of the original test, which the reviewer notes. assessment finding SEC-2025-03 was excluded from an earlier draft of TDR-2026-16 and reinstated after the reviewer obtained the supporting record.

Residual exposure on Low-severity finding SEC-2026-13 reads low. SEC-2026-13 produces this judgement: the finding was closed within the committed window and the closure evidence was inspected. Finding SEC-2026-13 is remediated, with a closure date of 2026-05-15. In Low-severity finding SEC-2026-13 the exploitation path means a large payload degrades broker throughput. Low-severity finding SEC-2026-13 was verified to the extent the available evidence permits, and the residual uncertainty is stated rather than absorbed.

8. Reliability debt and what it would cost to clear

Battery Management Service item TD-21 is sequenced behind a platform change that has its own target date. Register item TD-21 carries a 2026-09-30 target and a status of in progress. Funding for quality item TD-21 is partly funded, with the second phase unbudgeted, and the register does not say so. TD-21 on Battery Management Service is the entry that changed most between the first and second document requests behind TDR-2026-16.

Replace the fallback scan matcher with the current implementation is entered at TD-13 as an performance item. Effort on programme item TD-13 is estimated at 1.5 engineer-quarters, costed at $178,000. TD-13 on Localization Engine carries this observation: carries the same dependency as the copyleft removal. performance item TD-13 is not material on its own and belongs in this report because of what it reveals about the surrounding process.

Test capability item TD-05 is the item to bring the second bench rig to the current sensor configuration, classified under test capability. The reviewer's qualification on Hardware-in-the-Loop Rig item TD-05 is that outstanding since the April 2025 lidar timing incident. Test capability item TD-05 is partly funded, with the second phase unbudgeted; the register status and the funding position are not the same thing. TD-05 on Hardware-in-the-Loop Rig was excluded from an earlier draft of TDR-2026-16 and reinstated after the reviewer obtained the supporting record.

Quality item TD-04 is on the register because it defect escape rate on the fusion path is the highest in the fleet. Effort on register item TD-04 is estimated at 2 engineer-quarters, costed at $240,000. Funding for Costmap Fusion item TD-04 is not funded, and carried in the register regardless, and the register does not say so. Costmap Fusion item TD-04 appears twice in the working papers for TDR-2026-16 because the first answer was corrected after a follow-up request.

The dependency structure places Charge Scheduler item TD-23 behind a supplier release outside the company's control. Effort on TD-23 is estimated at 0.75 engineer-quarters, costed at $92,000. The consequence of leaving TD-23 on

Charge Scheduler undone is that it one noisy query affects task assignment. register item TD-23 is the clearest instance in TDR-2026-16 of a control that exists on paper and is weaker in practice.

The consequence of leaving TD-10 undone is that it the analytics load fails when the function is withdrawn. Register item TD-10 is sized at 0.5 engineer-quarters, or $54,000 at the company's loaded rate. Whether TD-10 on Analytics Warehouse Loader happens depends on funding, and it is unfunded and dependent on capital rather than engineering effort. Analytics Warehouse Loader item TD-10 rests on a single source, and the reviewer has said so rather than implying corroboration.

TD-19 matters because it the protective distance rests on a fixed approach speed assumption. The plan treats TD-19 as unfunded and dependent on capital rather than engineering effort, which the reviewer checked against the headcount model. As at the review date TD-19 on Speed and Separation Monitor was in progress, with 2026-08-31 still the stated target. programme item TD-19 was excluded from an earlier draft of TDR-2026-16 and reinstated after the reviewer obtained the supporting record.

TD-20 is funded in the current financial year; the register status and the funding position are not the same thing. Reading TD-20 on Wheel Odometry Filter against the funded plan, the prototype uses a network-copyleft component that must be replaced before shipping. The register shows TD-20 as in progress, due 2026-09-15. safety item TD-20 illustrates why TDR-2026-16 separates the severity of a finding from the confidence behind it.

Reading the programme plan, TD-06 is behind a supplier release outside the company's control. Not funded, and carried in the register regardless is the funding position on Simulation Harness item TD-06. Progress on quality item TD-06 is recorded as blocked against a 2026-11-30 date. programme item TD-06 is the only entry in its group where the reviewer and the engineering team reached different conclusions.

Sequencing of programme item TD-18 matters: it is behind a platform change that has its own target date. The estimate for TD-18 on Safety Controller is 0.75 engineer-quarters and $104,000. Programme item TD-18 is funded only if the team's headcount plan is met in the engineering plan the reviewer was shown. Safety Controller item TD-18 is the entry that changed most between the first and second document requests behind TDR-2026-16.

9. Incidents with a safety dimension

Hazard HZ-21 in the register applies to every robot variant currently in production. Hazard HZ-21 carries an open qualification: both currently shipping variants were tested. HZ-21 depends for its mitigation on push detection that engages the brake and requires a deliberate release. HZ-21 is the only entry in its group where the reviewer and the engineering team reached different conclusions.

Hazard HZ-08 in the register is not clean: the tie-break landed in October 2025 after two stall events. On history, HZ-08 has produced no event, though the control has activated repeatedly. Hazard HZ-08 in the register was scoped to the manipulator-equipped variant only, and the reviewer checked the scoping against the fleet record. Equipment damage hazard HZ-08 is not material on its own and belongs in this report because of what it reveals about the surrounding process.

Occasional-exposure hazard HZ-20 was verified through contamination bench test plus a review of cleaning compliance at three sites. Field history for Serious injury hazard HZ-20 is that it produced one event whose investigation is still open. The scope of register entry HZ-20 is the sites that have moved to opportunity charging. hazard HZ-20 in the register was discussed with two engineers who gave slightly different accounts, and the difference is preserved here.

Severity at hazard HZ-04 in the register is Serious injury, with the exposure rated Remote and controllability Low. Where HZ-04 bites is robots fitted with the weighing deck. Remote-exposure hazard HZ-04 has never activated in the field and is evidenced only on the bench, which is the strongest available test of the control. hazard HZ-04 in the register is included because a reader of TDR-2026-16 should be able to disagree with the inference without repeating the work.

Produced two near-miss reports from site staff is the incident history behind Frequent-exposure hazard HZ-01. Where Serious injury hazard HZ-01 bites is every site with a mezzanine or ramp. Serious injury hazard HZ-01 is not clean: the fixed approach speed assumption is carried as an open qualification. HZ-01 was tested rather than accepted, and the test is described in the working papers behind TDR-2026-16.

Exposure under hazard HZ-10 in the register extends to the newest three sites, where commissioning is incomplete. Evidence of verification at register entry HZ-10 consists of supplier test report plus an internal witnessed disconnect test. Produced a single event before the current control was fitted is the incident history behind hazard HZ-10. Fire hazard HZ-10 is the reason one section of TDR-2026-16 concludes narrowly rather than generally.

The register places at HZ-11 the scenario in which battery pack thermal runaway during opportunity charging. Verification of Fire hazard HZ-11 was by supplier abuse test report plus a review of pack fault records. The parameter set for HZ-11 reads Fire, Improbable, Low. hazard HZ-11 was discussed with two engineers who gave slightly different accounts, and the difference is preserved here.

The scope of hazard HZ-03 in the register is every robot variant currently in production. Hazard HZ-03 in the register carries residual risk of Tolerable where the weighing deck is fitted with a status of partially verified. One qualification on register entry HZ-03 the reviewer would not remove is that robots without the weighing deck rely on operator loading discipline. HZ-03 illustrates why TDR-2026-16 separates the severity of a finding from the confidence behind it.

Safety controller fault log overwrites evidence needed for an investigation is the hazard entered at No direct harm hazard HZ-17. The residual entry for Frequent-exposure hazard HZ-17 reads Tolerable, with an evidence-loss caveat, open. The design measure recorded at HZ-17 is periodic fault log extraction to the cloud with a retention policy. HZ-17 was cross-checked against the engineering record supplied for TDR-2026-16, and the two agree.

On history, HZ-14 produced two near-miss reports from site staff. Severity at hazard HZ-14 is Minor injury, with the exposure rated Occasional and controllability Low. After mitigation, hazard HZ-14 is Tolerable once the stale roster defect is closed; the register status is verified. hazard HZ-14 has an owner who accepted the finding immediately, which is worth recording alongside the finding itself.

10. Unresolved matters and management's declinatures

Three of the open items in the report at TDR-2026-16 would close on a single document that the company has agreed to produce. The reviewer has resisted the temptation to tidy report TDR-2026-16: the open items are open because the evidence is not there yet. Method in report TDR-2026-16 was three-fold: read the design record, interview the owning engineers, and verify the claim against a running system. In the report at TDR-2026-16 the reviewer distinguishes what was verified from what was represented, and marks each conclusion accordingly. The reliability record, incident history and post-incident follow-through of the fleet platform produced more evidence than TDR-2026-16 reproduces, and the working papers hold the remainder.

The report at TDR-2026-16 keeps declined items visible: risk acceptance without an expiry date is itself recorded as a finding. Report TDR-2026-16 treats a backlog label as evidence of intent and not as evidence of remediation, which is why several items remain open in this report that the company regards as handled. TDR-2026-16 addresses the reliability record, incident history and post-incident follow-through of the fleet platform. It does not address commercial terms, financial reporting or employment matters, which sit with other workstreams. 2026-04-27 is the evidence cut-off for this report, whose subject is the reliability record, incident history and post-incident follow-through of the fleet platform. R. Ostrowski has kept the working papers behind report TDR-2026-16 in a form that a second reviewer could re-perform.

Pinehurst Systems Assurance recommends that the 8 indicators be reported with their measurement basis attached, since two of them changed basis during the reviewed period. Monitoring recommended by the report at TDR-2026-16 relies on metrics the company already emits, so no new instrumentation is required to start. The report at TDR-2026-16 reports two findings the company disputes; the dispute is recorded next to the finding rather than resolved silently. The evidence convention in this report is that severity describes consequence and evidence strength describes confidence, and the two are never merged. Pinehurst Systems Assurance raised each finding with the owning engineer before it was written up, and the responses are recorded.

R. Ostrowski has left several concerns in this report at limited evidence strength rather than dropping them or dressing them up. Readers of this report should note that a moderate-evidence finding may carry a high severity, and that this is not a contradiction. The monitoring pack the report at TDR-2026-16 proposes would have surfaced two of the incidents examined here before a customer noticed. R. Ostrowski notes in this report that the company's own reporting is

adequate for three of the four recommended indicators. The evidence behind TDR-2026-16 is retained by Pinehurst Systems Assurance and is available to Ardent Peak on request.

11. Conclusions on reliability and post-close monitoring

For Ardent Peak's purposes, TDR-2026-16 concludes that the engineering claims that matter to the investment case are supportable, and two of the claims the company leads with are not. TDR-2026-16 concludes that the engineering claims that matter to the investment case are supportable, and two of the claims the company leads with are not. 11 follow-up items arose from the interview programme for the report at TDR-2026-16; 1 were still open at delivery. Report TDR-2026-16 draws on 9 interviews, including the engineering owners of every component examined in detail. R. Ostrowski has kept the company's terminology even where a different word would have read better, so that a reader can search the source documents.

The present report separates matters that bear on the closing conditions from matters that belong in the first two quarters after close. Nothing in the report at TDR-2026-16 supports a retention or an escrow; the findings are operational and they are funded. This report, the report at TDR-2026-16, is the Pinehurst Systems Assurance workstream on the reliability record, incident history and post-incident follow-through of the fleet platform, delivered to Ardent Peak on 2026-04-27. The instruction behind TDR-2026-16 was narrow: examine the reliability record, incident history and post-incident follow-through of the fleet platform and report by 2026-04-27. Where report TDR-2026-16 uses the word verified, it means the reviewer performed or witnessed something, and nothing weaker.

Post-close monitoring under the present report would track four things: objective attainment, change failure rate, open finding ageing and coverage on the control path. Reporting under TDR-2026-16 should distinguish a target that was met from a target that was lowered, which the current pack does not. The overall conclusion of report TDR-2026-16 is that the engineering claims that matter to the investment case are supportable, and two of the claims the company leads with are not. Stated without qualification for once, the conclusion of the report at TDR-2026-16 is that the engineering claims that matter to the investment case are supportable, and two of the claims the company leads with are not. Nothing in report TDR-2026-16 is intended as a legal opinion or as advice on the adequacy of insurance cover.